Satellite Data in Agriculture: A High-Level Summary

Current uses, soil-health inference, and what the evidence supports | October 2026

1. Why satellites matter for agriculture

Agriculture must deliver food, fibre and fuel for a growing population while adapting to climate change and protecting soil, water and biodiversity. Earth-observation (EO) satellites help because they provide repeated, spatially complete, objective measurements of the land surface at scales from individual fields to continents. The canonical meta-review of the field (Weiss, Jacob & Duveiller, 2020, *Remote Sensing of Environment*) frames the value of remote sensing as supplying repetitive information on crop status throughout the season, at different scales, for different actors - from breeders and farmers to insurers, governments and food-security agencies.

Three developments over the past decade turned EO from a research tool into operational infrastructure: (i) free and open data policies (Landsat since 2008; the Copernicus Sentinels since 2014-15), (ii) sensors with field-relevant resolution and frequent revisit (Sentinel-2's 10-20 m pixels, red-edge bands and ~5-day revisit), and (iii) cloud platforms such as Google Earth Engine and the Copernicus Data Space Ecosystem that remove the need to download and store petabytes of imagery.

2. What satellites are used for today

Application area	What is measured / inferred	Typical data and methods	Maturity
Cropland and crop-type mapping	Where crops grow, which crop, field boundaries, cropping intensity	Sentinel-2 / Landsat / Sentinel-1 time series; random forest, gradient boosting, deep learning; e.g. USDA Cropland Data Layer, ESA WorldCereal	Operational
Crop condition and phenology	Greenness (NDVI/EVI), leaf area index, chlorophyll and nitrogen status, emergence, peak, senescence	Vegetation indices and red-edge indices; PROSAIL radiative-transfer inversion; time-series smoothing	Operational
Yield estimation and forecasting	Field to national yield, biomass, anomalies vs. previous seasons	Index regression, ML/DL on image time series, assimilation into crop models (WOFOST, DSSAT, APSIM) with weather and soil data	Operational at regional scale; field scale maturing
Precision / site-specific management	Within-field variability for variable-rate N, seeding, irrigation, management zones	High-resolution optical (Sentinel-2, PlanetScope), NDRE / chlorophyll indices, zoning algorithms	Commercial products widespread
Water and drought	Soil moisture, evapotranspiration (ET), crop water stress, irrigation extent	SMAP / SMOS passive microwave, Sentinel-1 SAR, Landsat / ECOSTRESS thermal, energy-balance ET models (e.g. FAO WaPOR)	Operational
Soil properties	Topsoil organic carbon, clay, carbonates, pH, cation exchange capacity, bulk density proxies	Bare-soil composites from Sentinel-2 / Landsat; PLSR, Cubist, random forest, deep learning calibrated on soil samples (LUCAS, national surveys); digital soil mapping with hundreds of covariates (SoilGrids)	Research to early operational; moderate accuracy

Application area	What is measured / inferred	Typical data and methods	Maturity
Soil biology (emerging)	Microbial biomass, bacterial and fungal diversity, fungal:bacterial balance	Statistical links between NDVI / hyperspectral imagery and eDNA or PLFA ground data; Gaussian-process and random-forest models	Exploratory; indirect inference only
Policy, insurance and compliance	CAP subsidy checks, cover-crop and tillage practice detection, disaster assessment, carbon MRV	Sentinel-1/2 time series, change detection, practice classifiers	Operational in EU; growing elsewhere

3. Soil fertility and soil health from orbit: what is - and is not - possible

Directly sensed chemistry. Soil organic carbon (SOC), iron oxides, carbonates and clay minerals have characteristic absorption features in the visible, near-infrared and shortwave-infrared (400-2500 nm). When soil is bare, dry and relatively smooth, multispectral (Sentinel-2, Landsat) or hyperspectral (EnMAP, PRISMA, EMIT) reflectance can be regressed against laboratory values. Reviews of the SOC literature (Angelopoulou et al., 2019; Lima et al., 2025) report that Sentinel-2 and Landsat 8 are now the dominant data sources, that vegetation and topographic indices are among the most influential predictors, and that the spatial spread of calibration samples matters more than their total number. Published satellite SOC models typically reach cross-validated R^2 of about 0.4-0.7; PLSR models from Sentinel-2 bare-soil spectra achieve intermediate skill ($R^2 \geq 0.5$, RPD ≥ 1.4) for SOC, pH, $CaCO_3$ and CEC in favourable European settings.

Indirectly mapped properties. pH, total nitrogen and CEC have weak or no direct spectral signature; they are predicted because they co-vary with SOC, clay, carbonates, climate, terrain and parent material. This is the logic of digital soil mapping: SoilGrids 2.0 (Poggio et al., 2021) uses machine learning on ~240,000 soil profiles and 400+ environmental covariates (many satellite-derived) to map SOC, total N, pH, CEC, bulk density and texture at 250 m globally, with spatially explicit 5th-95th percentile uncertainty.

Soil biology. Microbial biomass and the fungal:bacterial (F:B) ratio cannot be observed from space. Recent work shows that satellite data can be *statistically associated* with soil biology: hyperspectral imagery from DESIS explained up to about half of the variance in forest soil bacterial alpha diversity and ~40% for fungi when paired with eDNA data; and Sentinel-2 NDVI, after adjusting for soil and climate confounders, has been linked to the abundance of beneficial versus pathogenic fungal genera in European cereal fields. These results are promising but exploratory, and F:B ratios still require field measurement (PLFA, qPCR, microscopy-based kits).

4. How the data get from orbit to a decision

- **Acquire:** optical multispectral, hyperspectral, thermal, radar (SAR) and passive-microwave sensors each capture a different physical property of crops and soil.
- **Pre-process:** calibration, atmospheric correction to surface reflectance, cloud masking, geometric alignment and temporal compositing produce analysis-ready data.
- **Derive features:** vegetation and soil indices, biophysical variables, phenology metrics and bare-soil composites.
- **Model:** statistical, machine-learning or process-based models link features to ground truth (yields, crop labels, soil analyses).
- **Evaluate and deliver:** spatial cross-validation, independent test sets, uncertainty quantification, and translation into maps, forecasts and prescriptions.

5. Key take-aways for practitioners

- Satellite data excel at **spatial coverage, revisit frequency and consistency**; they complement rather than replace ground sampling.
- Sentinel-2 plus Sentinel-1 (free, 10-20 m, 5-6 day revisit) is the default starting point for most agricultural work; Landsat gives the long archive; MODIS gives daily regional context.
- For **soil properties**, expect moderate accuracy, strong dependence on bare-soil availability and local calibration, and much better results for SOC/clay/carbonates than for pH, N or CEC.
- For **soil biology**, satellite products are indicators of habitat and plant response, not measurements of microbial communities.
- Always report validation metrics and uncertainty, and check whether a model transfers across years, regions and sensors before operational use.

Key sources: Weiss, Jacob & Duveiller (2020) Remote Sens. Environ. 236:111402 - Angelopoulou et al. (2019) Remote Sens. 11:676 - Lima et al. (2025) Remote Sens. 17:882 - Poggio et al. (2021) SOIL 7:217-240 - Gorelick et al. (2017) Remote Sens. Environ. 202:18-27 - Segarra et al. (2020) Agronomy 10:641 - Cardille et al. (eds., 2024) Cloud-Based Remote Sensing with Google Earth Engine (Springer, open access). Full reference list with DOIs is provided in the companion document *Foundational and Canonical References*.